```
PS D:\PYTHON\Project-4> & C:\Users\Dhruvi\AppData\Local\Programs\Python\Python314\python.exe d:/PYTHON/Project-4/Functional_Treat_project.py
WELCOME to the DATA ANALYZER and TRANSFORMER:)

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:1

Enter data for 1D array (separated by spaces): 34 56 78 24 90 13 16 45
Data has been stored successfully!

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:2

DATA SUMMARY:
-Total elements: 8
-Minimum Value: 13
-Maximum Value: 90
-Sum of all values: 356
-Average Value: 44.5

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
```

```
Enter your choice here:3

Enter a number to calculate its factorial:6
Factorial of 6 is: 720

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:4
Enter a threshold value to filter out data above this value:
60

Filtered data (value >= 60)
78, 90

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:5

Choose Sorting option:
1. Ascending
2. Descending

Enter your choice:1

Sorted Data in Ascending Order:
13, 16, 24, 34, 45, 56, 78, 90

Main Menu:
1. Input Data
```

```
Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:5

Choose Sorting option:
1. Ascending
2. Descending

Enter your choice:2

Sorted Data in Descending Order:
90, 78, 56, 45, 34, 24, 16, 13

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:6

Dataset Statistics:
-Minimum Value: 13
-Maximum Value: 90
-Sum of all values: 356
-Average Value: 44.5

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
```


```
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:5

Choose Sorting option:
1. Ascending
2. Descending

Enter your choice:2

Sorted Data in Descending Order:
90, 78, 56, 45, 34, 24, 16, 13

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
Enter your choice here:6

Dataset Statistics:
-Minimum Value: 13
-Maximum Value: 90
-Sum of all values: 356
-Average Value: 44.5

Main Menu:
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit
```


```
Enter your choice here:7
Exiting The program. Thank you for using this program. Goodbye!
PS D:\PYTHON\Project-4>
```